

MaxwellFD: 2D Yee FDTD and FDFD

Physics, methods, and code

Tom Smy

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Abstract

MaxwellFD is a Python Yee-cell solver. One uniform Cartesian grid, one material sampling, and one pair of discrete curls feed both a leapfrog (FDTD) and a sparse frequency-domain system (FDFD). Phasors use $e^{j\omega t}$, the same convention as MaxwellMOM. This note describes the physics and the method that are actually coded, how the package is organized, and how the Fresnel-slab and dielectric-cylinder examples are run and drawn. The equation reference, including the PML, the resistive sheet, and the GSTC cell, is `FD_LaTeX_Reference.tex`. Those three items are specified there and are not in the operators.

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1 Status

Version 0.1.0 contains two slices.

1. The 2D cell in free space and in a uniform isotropic medium: TMz and TEz curls, PEC and periodic walls, the CFL limit, cavity eigenvalues, a manufactured FDFD solve, and a periodic standing wave.
2. Volumetric media. A staircase assigns a real ε , μ , and $\sigma \geq 0$ to each Yee sample. FDFD accepts a complex $\varepsilon(\omega)$. FDTD adds one bulk Debye pole or one bulk Lorentz pole through a polarization current. The closed-box checks are a Fresnel slab and a dielectric cylinder, both driven by an analytic Dirichlet trace on the existing PEC wall.

MaxwellFD does not import MaxwellMOM. The shared pieces are the time convention, the four SI constants, and the analytic targets used as external checks. There is no 3D grid, no total-field/scattered-field boundary, no embedded PEC or PMC object, and no sheet unknown.

2 Physics

Fields are in SI units. With the convention $e^{j\omega t}$, a wave $e^{-jk\hat{\mathbf{k}}\cdot\mathbf{r}}$ travels in the direction $\hat{\mathbf{k}}$, and ∂_t becomes $j\omega$. The constants in `maxwell_fd.utils.constants` are

$$\begin{aligned} c_0 &= 299792458 \text{ m s}^{-1}, & \varepsilon_0 &= 8.8541878128 \times 10^{-12} \text{ F m}^{-1}, \\ \mu_0 &= 1.25663706212 \times 10^{-6} \text{ H m}^{-1}, & \eta_0 &= 376.73031346177 \Omega. \end{aligned}$$

Faraday's law and Ampere's law with an impressed current $\mathbf{J}$ are

$$\nabla \times \mathbf{E} = -j\omega\mu\mathbf{H}, \quad (1)$$

$$\nabla \times \mathbf{H} = j\omega\varepsilon\mathbf{E} + \mathbf{J}. \quad (2)$$

A real conductivity is not a second impressed current. In FDFD it enters as $\varepsilon + \sigma/(j\omega)$ when ε is real. A law that already returns a complex $\varepsilon(\omega)$ has included its loss, and the accompanying σ must be zero.

The grid lies in the xy -plane and nothing depends on z . TMz keeps E_z , H_x , and H_y . TEz keeps H_z , E_x , and E_y . The code never forms the scalar Helmholtz operator. Both drivers eliminate, or update, the magnetic field through the curls, so a face-by-face μ stays in the magnetic samples.

On a PEC wall the tangential electric field is the boundary value stored with the grid. Milestone 1 sets that value to zero. The slab and the cylinder instead prescribe the analytic tangential trace and leave the impressed current at zero. A periodic wall identifies opposite faces with phase one. PMC, a stretched-coordinate PML, and a resistive or GSTC sheet are written in the formulation note and are not boundary kinds in `YeeGrid2D`. The grid accepts `Boundary.PEC` and `Boundary.PERIODIC` only.

2.1 Staircase media

A sample is inside a region when its own coordinate is inside. Electric and magnetic samples are tested separately. There is no volume fraction and no conformal cut. A slab perpendicular to y occupies the half-open interval $y_0 \leq y < y_1$. A disk of radius R centered at (x_c, y_c) occupies $(x - x_c)^2 + (y - y_c)^2 \leq R^2$. Later regions overwrite earlier ones. The background is the law passed to `StaircaseIsotropic`, usually `UniformIsotropic` with vacuum values.

2.2 Bulk dispersion

One isotropic electric pole may fill the whole domain. The staircase does not yet vary the pole in space, and there is no magnetic dispersion.

The Debye permittivity and its auxiliary are

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{\varepsilon_0 \Delta \varepsilon}{1 + j\omega\tau}, \quad (3)$$

$$\tau \partial_t P + P = \varepsilon_0 \Delta \varepsilon E. \quad (4)$$

The Lorentz permittivity and its auxiliary are

$$\varepsilon(\omega) = \varepsilon_\infty + \frac{\varepsilon_0 \omega_p^2}{\omega_0^2 - \omega^2 + j\gamma\omega}, \quad (5)$$

$$\partial_t^2 P + \gamma \partial_t P + \omega_0^2 P = \varepsilon_0 \omega_p^2 E. \quad (6)$$

These are bulk oscillators. They are not the surface oscillators in the GSTC section of the formulation note. Ampere's update subtracts the polarization current J_p with the impressed current,

$$\mathbf{E} \leftarrow C_a \mathbf{E} + C_b (\nabla \times \mathbf{H} - \mathbf{J}_p - \mathbf{J}). \quad (7)$$

C_a and C_b are the usual lossy coefficients built on ε_∞ and σ . The Lorentz auxiliary stores P at integer steps and advances it explicitly from E^n . The Debye auxiliary is centered, $E^{n+1/2} = (E^{n+1} + E^n)/2$, and the update is local to each sample. TEz keeps one auxiliary on E_x and one on E_y . A complex ε is rejected by the leapfrog. Passing ω into `sample_tm` or `sample_te` is how FDFD sees (3) or (5) directly.

3 Discrete method

3.1 Grid

The domain is $[0, n_x \Delta x] \times [0, n_y \Delta y]$. Array axis 0 is x and axis 1 is y , so a sample is indexed $[i, j]$. Locations are those of the Yee cell. For a PEC grid the boundary electric samples are stored. TMz places E_z on nodes, H_x on the y -edges, and H_y on the x -edges. TEz places H_z at cell centers, E_x on y -edges, and E_y on x -edges. E_x is zero on $y \in \{0, b\}$ and E_y is zero on $x \in \{0, a\}$ unless a Dirichlet trace replaces those zeros. Periodic identification drops the duplicate far side, and every stored sample is an unknown.

`YeeGrid2D.coordinates` returns the 1D sample coordinates of one component. Those coordinates are the positions the staircase tests.

3.2 Curls

The discrete operators are centered differences. On the PEC grids they are transposes of each other once the boundary electric samples have been removed from the unknown vector. An invalid degree-of-freedom id is -1 and must be masked before any component offset is added. TMz uses

$$(\nabla \times E)_x = \partial_y E_z, \quad (\nabla \times E)_y = -\partial_x E_z, \quad (8)$$

and the matching Ampere pair. TEz uses

$$\nabla \times \mathbf{E} = \partial_x E_y - \partial_y E_x, \quad E_x = \partial_y H_z, \quad E_y = -\partial_x H_z. \quad (9)$$

The array curls and the sparse curls implement the same stencil. The array form keeps every sample, including the boundary values. The sparse pair used by FDFD drops the PEC rows.

The sign in (1) is not stored inside the factored FDFD matrix. Reconstructing a physical magnetic field from a Yee electric field requires

$$\mathbf{H} = -\frac{\nabla \times \mathbf{E}}{j\omega\mu}. \quad (10)$$

Omitting the minus negates H_x or H_z . The example plots use (10).

3.3 Leapfrog

One step advances $\mathbf{H}$ from time $n - \frac{1}{2}$ to $n + \frac{1}{2}$, then $\mathbf{E}$ from n to $n + 1$. With real ε and σ ,

$$C_a = \frac{1 - \sigma\Delta t/(2\varepsilon)}{1 + \sigma\Delta t/(2\varepsilon)}, \quad C_b = \frac{\Delta t/\varepsilon}{1 + \sigma\Delta t/(2\varepsilon)}. \quad (11)$$

$J = 0$ and $J_p = 0$ is the milestone-1 update. The stable step on a uniform medium of speed c satisfies

$$\Delta t \leq \frac{1}{c\sqrt{\Delta x^{-2} + \Delta y^{-2}}}. \quad (12)$$

When $\sigma = 0$ the leapfrog invariant, not the sum of the staggered sample energies, is the conserved quantity. The fully discrete frequency of a mode lies slightly above its semi-discrete frequency.

3.4 FDFD and the Dirichlet trace

Eliminating $\mathbf{H}$ produces the electric system

$$\nabla \times (\mu^{-1} \nabla \times \mathbf{E}) - \omega^2 \varepsilon \mathbf{E} = -j\omega \mathbf{J}, \quad (13)$$

stored as $Ae = -j$. The real eigenproblem $Ke = \omega^2 Me$ still requires real positive ε and μ and $\sigma = 0$. On a TEz PEC box the curl-curl operator has a gradient kernel, so a computed ω at roundoff is discarded before the physical spectrum is read.

Boundary electric samples stay out of the unknown vector. A trace e_D that equals the prescribed boundary values, and is zero on the free samples, moves to the right-hand side,

$$A_{ff} e_f = -j - (Ae_D)_f. \quad (14)$$

The free-row action is the array form of the left-hand side of the FDFD equation. Calling `solve` with no trace, or with a zero trace, is the milestone-1 solve. The slab and the cylinder pass `dirichlet` set to the analytic boundary field and set $j = 0$. This closes the box with a prescribed trace. It does not stretch a coordinate and it does not add auxiliary fields.

3.5 Analytic checks

Both scattering checks use the trace (14) on every PEC electric sample. The interior material is the staircase of Sec. 2.1. The comparison is the gap left by numerical dispersion and, for the cylinder, by the staircase.

Fresnel slab. A non-magnetic slab $y_1 \leq y < y_2$ of index $n = \sqrt{\varepsilon_r}$ sits in vacuum. The incident wave travels toward $+y$ and has phase zero at the front face. TMz reflects with $r = (1-n)/(1+n)$. TEz reflects H_z with $r_H = (n-1)/(n+1) = -r$, so $R_H = -R$ while the transmission amplitude, including the transit phase e^{-jkd} , is the same. The interior amplitudes are not the same: TMz splits $1 \pm R$ with $1/n$, and TEz splits $1 \pm R_H$ with n . When $\varepsilon_r = 1$, both reflections vanish and the field is the incident wave, not a transmission of 1. The closed forms are `slab_coefficients` and `slab_fields`. A forced branch (`below`, `slab`, `above`) evaluates one piecewise piece at a face, where the two neighboring expressions must agree.

The comparison wavelength is kept off the TEz eigenfrequencies of the PEC box. On the 8×80 domain with $\Delta = 1$, $\lambda = 40$ is one of those frequencies and the Dirichlet matrix is singular. The shipped example uses $\lambda = 46$, $\Delta = 0.5$, and faces $y = 30.375$ and $y = 50.375$, which miss the electric nodes of both that grid and the once-refined grid.

Dielectric cylinder. The outgoing wave is $H_n^{(2)}$. The incident field e^{-jkx} has the Jacobi–Anger coefficients $(-j)^n$. TMz is the E_z series. TEz is the H_z series, with the contrast on the undifferentiated interior Bessel function, because E_φ brings $1/\varepsilon$. The PEC coefficient $a_n = -(-j)^n J_n(ka)/H_n^{(2)}(ka)$ is a separate regression. It is not the limit $\varepsilon_r \rightarrow \infty$ of the dielectric formula. The truncation follows the MaxwellMOM rule $\lceil |ka| + 4|ka|^{1/3} + 8 \rceil$, and the derivative is the recurrence $\frac{1}{2}(f_{n-1} - f_{n+1})$.

4 Code

The installable package is `maxwell_fd`, version 0.1.0. Its source is `src/maxwell_fd`. NumPy and SciPy are required. Matplotlib is the `viz` extra and is also a dependency of the conda environment `maxwell-fd`.

Module	Role
<code>grid.yee2d</code>	YeeGrid2D, Polarization, Boundary, coordinates
<code>materials.volume</code>	Uniform, staircase, mapped $\varepsilon(x, y)$, component arrays
<code>materials.dispersion</code>	DebyeDielectric, LorentzDielectric, and their auxiliaries
<code>operators.curl2d</code>	Sparse <code>curl_e</code> , <code>curl_h</code> , array curls, DOF layout
<code>drivers.fdt2d</code>	<code>step_tmz</code> , <code>step_tez</code> , CFL step, leapfrog invariant
<code>drivers.fdfd2d</code>	FDFDOperator: system, eigenproblem, Dirichlet solve
<code>analytics.cavity2d</code>	Semi-discrete cavity frequencies and mode shapes
<code>analytics.dispersion</code>	Fully discrete frequency and the three-sample fit
<code>analytics.fresnel</code>	Slab coefficients and fields
<code>analytics.mie2d</code>	Cylinder coefficients and E_z , H_z , E_x , E_y

A frequency-domain TMz solve with a staircase disk has this shape. The trace argument is omitted when the wall is a homogeneous PEC.

```
grid = YeeGrid2D(nx, ny, dx, dy, Polarization.TMZ, Boundary.PEC)
law = StaircaseIsotropic(
    UniformIsotropic(),
    (Disk(cx, cy, radius, eps=eps_r * EPS0),),
)
operator = FDFDOperator(grid, law.sample_tm(grid))
ez = operator.solve(omega, np.zeros(operator.layout.n_e),
    dirichlet={"ez": ez_trace})
```

`sample_tm(grid)` and `sample_te(grid)` without ω return the real instantaneous coefficients used by the leapfrog (ε_∞ when the law is dispersive). Passing ω allows a complex permittivity. `step_tmz` and `step_tez` take the impressed current and the polarization current as optional arguments. Their defaults reproduce the milestone-1 update.

The real eigenproblem is `spatial_operator`, which returns the pair (K, M) . TEz results should drop nonpositive eigenvalues before they are compared with the semi-discrete formula.

5 Examples

Scripts under `examples/fdfd/2d/` and `examples/fdtd/` are the runnable demos. Placeholders under `examples/fdfd/3d` and `examples/fdtd` hold the directory tree and do not solve a 3D or a time-domain scattering problem. From the project directory, with the conda environment `maxwell-fd` active,

```
python examples/fdfd/2d/fresnel_slab.py
python examples/fdfd/2d/dielectric_cylinder.py
```

Each script writes a sibling directory whose name is the script stem plus the parameters of that run, in the order they are passed to `examples/run_dir.output_dir`. Floats are formatted with twelve significant digits and trailing zeros removed, so 2.0 becomes 2 and 0.5 stays 0.5. Keys do not contain underscores.

Field maps use `pcolormesh` with Gouraud shading. Each polarization is a row of three columns: the Yee field, the analytic field, and their difference. The cylinder columns are FD, Mie, and FD–Mie. The slab columns are FD, Fresnel, and FD–Fresnel. Color limits of the first two columns match. The difference has its own scale. The cylinder outline, or the slab faces, is drawn on every panel.

5.1 Fresnel slab

```
python examples/fdfd/2d/fresnel_slab.py \
    --index 2 --thickness 20 --y1 30.375 --wavelength 46 \
    --length 8 --height 80 --dx 0.5 --pml 1
```

`-index` is the refractive index n , and the permittivity used in the solve is $\varepsilon_r = n^2$. The default $n = 2$ is $\varepsilon_r = 4$. `-length` and `-height` are the region sides along x and y . Both must be a positive integer number of cells of size `-dx`, and each direction needs at least two cells. `-pml` is a thickness at each end in y only. It must be a positive multiple of Δ and must leave the slab in the open interval $(\text{pml}, \text{height} - \text{pml})$.

The default run writes

```
fresnel_slab_index2_d20_lam46_y130.375_length8_height80_dx0.5_pml1/
```

- `domain.png`, the mesh, the slab, the y -end bands, and the x walls;
- `fresnel_slab.png`, TMz E_z and TEz H_z in three columns;
- `cuts.png`, the center-column cuts of E_z , H_x , E_x , and H_z against the analytic curves.

On this grid the free-sample electric L^2 errors annotated on the figures are 0.23% (TMz) and 1.30% (TEz). Those numbers are the result of this particular grid. The acceptance bars in the formulation note are wider and were not changed to match them.

Propagation is $+y$, drawn horizontal so the figure reads with the line cuts. The incident field does not depend on x , and neither does the slab. The orange bands are therefore only the two ends $y = 0$ and $y = b$. The walls $x = 0$ and $x = a$ are drawn in green and labeled as the PEC trace: every boundary electric sample on those walls is set to the analytic Fresnel value, together with the samples on $y = 0$ and $y = b$. The slab rectangle spans the full width in x . Figure 1 is that layout. Nothing in `FDFDOperator` stretches a coordinate. The orange bands document where a later PML would close the propagation direction. They are not absorbing layers in this solve.

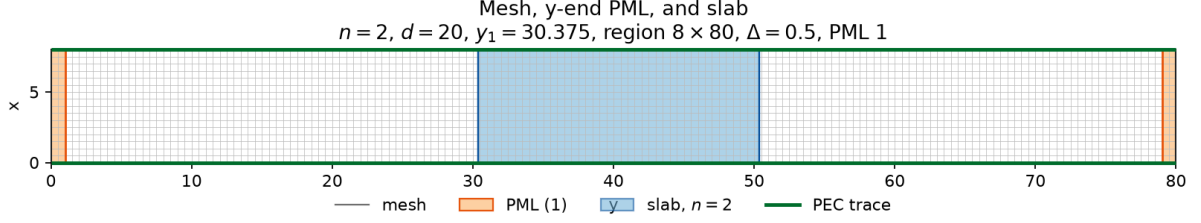


Figure 1: Default Fresnel domain. Orange bands are the y -end PML thickness drawn on the figure. Green edges are the PEC trace at $x = 0$ and $x = 80$.

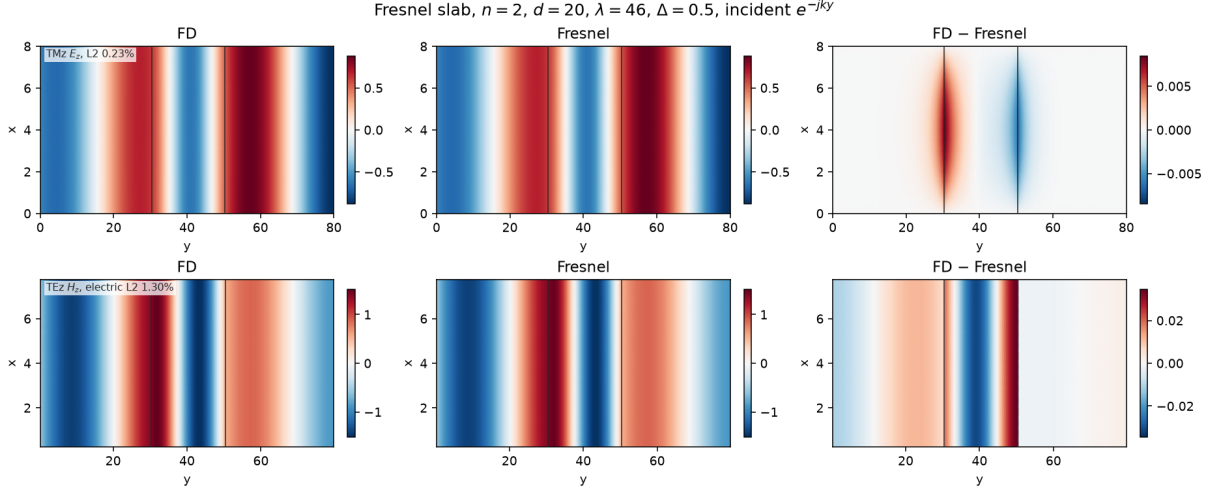


Figure 2: Default Fresnel maps. Vertical lines mark the slab faces. H_z is rebuilt with (10). The annotated L^2 errors are the free electric samples.

5.2 Dielectric cylinder

```
python examples/fdfd/2d/dielectric_cylinder.py \
    --radius 10 --index 1.5 --size 50 --dx 1 --pml 8
```

The region is a square of side `-size`. `-index` is again n , with $\varepsilon_r = n^2$. The default $n = 1.5$ is $\varepsilon_r = 2.25$. The size parameter ka is held at 1, so $k = 1/a$ and the wavelength scales with the radius. The radius must be positive and smaller than half the side length minus the PML thickness. The default run writes

```
dielectric_cylinder_index1.5_ka1_a10_size50_dx1_pml8/
```

with `domain.png` and `dielectric_cylinder.png`.

The cylinder scatters in the plane of the grid, so the domain figure draws the PML thickness on all four sides. As in the slab example, that frame is a layout. The solve is still (14) on the PEC wall, with the Mie field as the trace and with the staircase disk as the only material contrast. TMz shows $\text{Re } E_z$. TEz shows $\text{Re } H_z$ from (10), and the annotated L^2 error is the free-sample error in E_x and E_y . On the default grid those errors are 0.44% and 2.07%.

6 Tests and later slices

The suite is `pytest` from the project root, twenty-five tests in `tests/`. Milestone 1 checks the curls, the adjoint, the cavity eigenvalues, the manufactured solve, the invariant and the CFL growth, and the fully discrete dispersion, with relative or absolute bars from 10^{-12} to 10^{-6} as

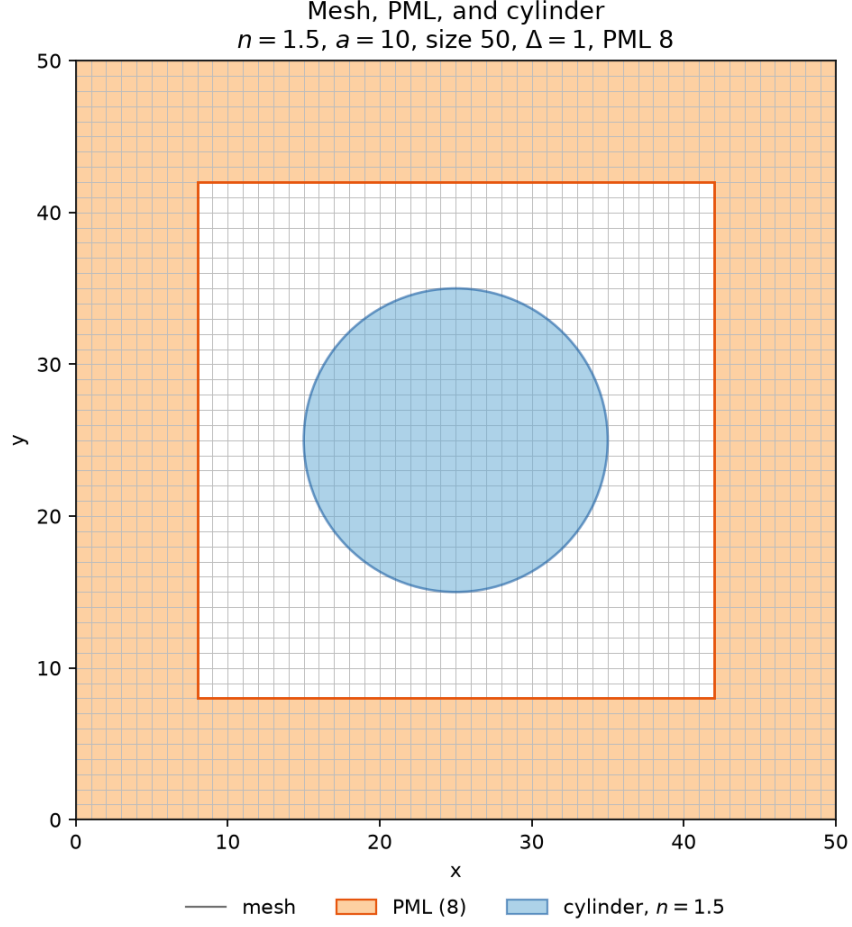


Figure 3: Default cylinder domain. The orange frame is the drawn PML thickness. The disk is the staircase geometry. The outer boundary of the solve remains the PEC trace.

listed in the formulation note. Milestone 2 adds the manufactured solve with $\sigma > 0$ and with a smooth $\varepsilon(x, y)$, a uniform $\varepsilon_r = 4$ cavity, the slab-loaded cavity, the lossless staircase invariant, Fresnel face continuity, the Mie continuity and PEC coefficient, the Fresnel and cylinder L^2 bars of 2×10^{-2} , and one period of Debye and Lorentz evolution. A failing bar is a reason to refine the grid in that test, not to widen the bar.

The formulation note already fixes the formulas for the slices that are not coded: the convolutional PML, embedded PEC and PMC, open-region Mie, the 3D cell, a resistive sheet, the symmetric GSTC cell, impressed beams, a near-to-far integral, and a Bloch phase. The example domain figures are the picture of where the y -end PML, or the four-sided PML, will sit. They do not implement the stretch.

References

- [1] K. S. Yee, “Numerical solution of initial boundary value problems involving Maxwell’s equations in isotropic media,” *IEEE Trans. Antennas Propag.*, 1966.
- [2] T. J. Smy, S. Stewart, and S. Gupta, “Integrated generalized sheet transition conditions in a Yee-cell based FDTD simulation of electromagnetic metasurfaces,” arXiv:1706.10136, 2017.

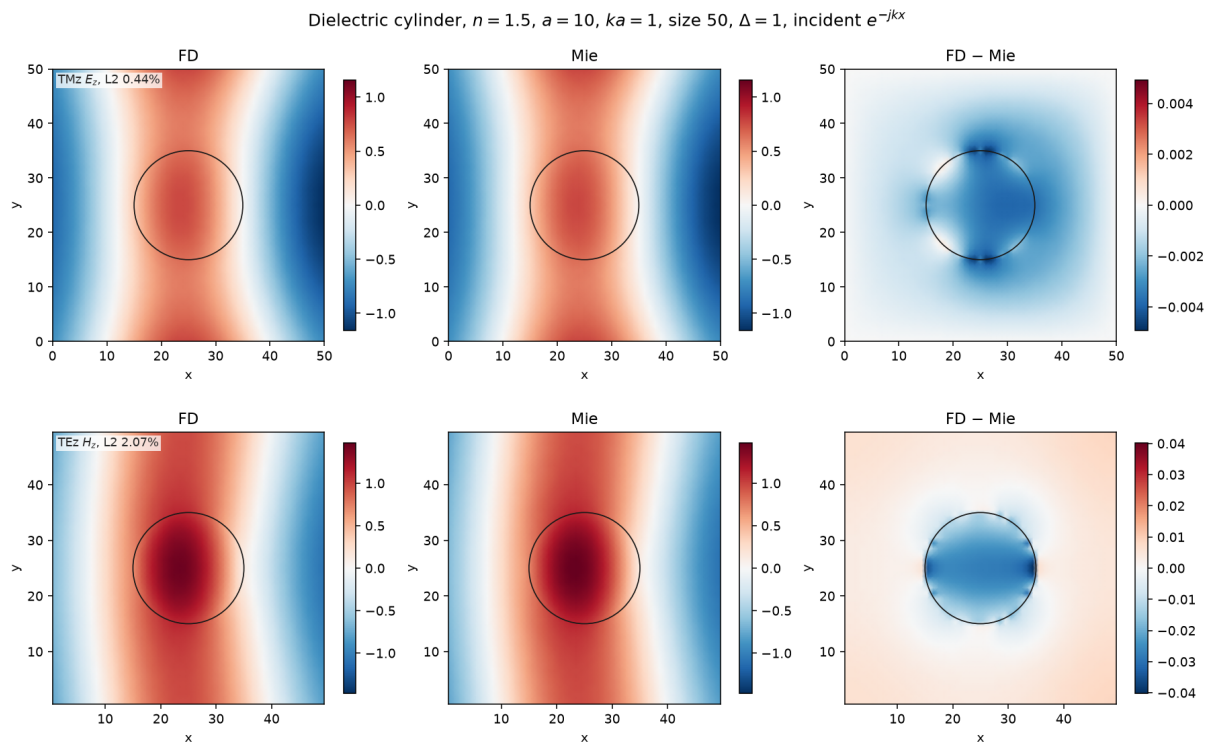


Figure 4: Default cylinder maps against the Mie series at $ka = 1$, $n = 1.5$.